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STAT525 HOMEWORK#8

DUE NOV 7, 2018, **BEFORE** CLASS

1. KNNL Problem 17.2
2. KNNL Problem 17.3
3. KNNL Problem 17.12 For d, please calculate the confidence intervals used for the plot but ignore the plot itself.
4. KNNL Problem 17.17 (Bonferroni adjustment is required for part b)

   Hint: you can use the following statement for $(1-\alpha)$ confidence interval of linear combination of $\sum c_i\mu_i$'s

   ```
   proc glm data=***;
   class machine;
   model y=machine / clparm alpha;
   estimate '**' machine c1 c2 c3 c4;
   ```

5. KNNL Problem 17.25 (Bonferroni adjustment is required)
6. KNNL Problem 18.2
7. KNNL Problem 18.17 part b to part d.

1. KNNL Problem 17.2

17.2. A trainee examined a set of experimental data to find comparisons that "look promising" and calculated a family of Bonferroni confidence intervals for these comparisons with a 90 percent family confidence coefficient. Upon being informed that the Bonferroni procedure is not applicable in this case because the comparisons had been suggested by the data, the trainee stated: "This makes no difference. I would use the same formulas for the point estimates and the estimated standard errors even if the comparisons were not suggested by the data." Respond.

The trainee is wrong.
The Bonferroni procedure can only be used when the set of comparisons is planned before looking at the data. Once the trainee looks at the results and then chooses the "promising" comparisons, this becomes a data-driven (post hoc) selection.

Even though the same point estimates and standard errors can be used, the confidence level is no longer valid. The Bonferroni method assumes the family of comparisons is fixed in advance. When the comparisons are chosen after seeing the data, the actual familywise confidence level will be much lower than 90%, which increases the chance of a Type I error (false positives).

If the comparisons are suggested by the data, the correct method to use is the Scheffé procedure, which is valid for all possible contrasts that might be chosen after looking at the data.

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2. KNNL Problem 17.3

17.3. Consider the following linear combinations of interest in a single-factor study involving four factor levels:

$$\text{(i)} \quad \mu_1 + 3\mu_2 - 4\mu_3$$

$$\text{(ii)} \quad .3\mu_1 + .5\mu_2 + .1\mu_3 + .1\mu_4$$

$$\text{(iii)} \quad \frac{\mu_1 + \mu_2 + \mu_3}{3} - \mu_4$$

a. Which of the linear combinations are contrasts? State the coefficients for each of the contrasts.

i) Coefficients $(1, 3, -4, 0)$ sum: $1+3-4+0=0$ Constrast ✓

ii) Coefficients $(0.3, 0.5, 0.1, 0.1)$ sum: $0.3+0.5+0.1+0.1=1 \neq 0$ not constrast

iii) Coefficients $(1/3, 1/3, 1/3, -1)$ sum: $1/3+1/3+1/3-1=0$ Constrast ✓

b. Give an unbiased estimator for each of the linear combinations. Also give the estimated variance of each estimator assuming that $n_i \equiv n$.

For $L = \sum c_i \mu_i$

• Estimator : $\hat{L} = \sum c_i \bar{Y}_i$ (unbiased)

• Estimator : $\widehat{Var}(\hat{L}) = MSE \sum c_i^2 / n = MSE/n \sum c_i^2$

i) $c = (1, 3, -4, 0)$

• $\hat{L} = \bar{Y}_1 + 3\bar{Y}_2 - 4\bar{Y}_3$

• $\sum c_i^2 = 1 + 9 + 16 + 0 = 26$

• $\widehat{Var}(\hat{L}) = 26/n \cdot MSE$

ii) $c = (0.3, 0.5, 0.1, 0.1)$

• $\hat{L} = 0.3\bar{Y}_1 + 0.5\bar{Y}_2 + 0.1\bar{Y}_3 + 0.1\bar{Y}_4$

• $\sum c_i^2 = 0.09 + 0.25 + 0.01 + 0.01 = 0.36$

• $\widehat{Var}(\hat{L}) = 0.36/n \cdot MSE$

iii) $c = (1/3, 1/3, 1/3, -1)$

• $\hat{L} = (\bar{Y}_1 + \bar{Y}_2 + \bar{Y}_3)/3 - \bar{Y}_4$

• $\sum c_i^2 = 1/9 + 1/9 + 1/9 + 1 = 4/3$

• $\widehat{Var}(\hat{L}) = (4/3n) \cdot MSE$

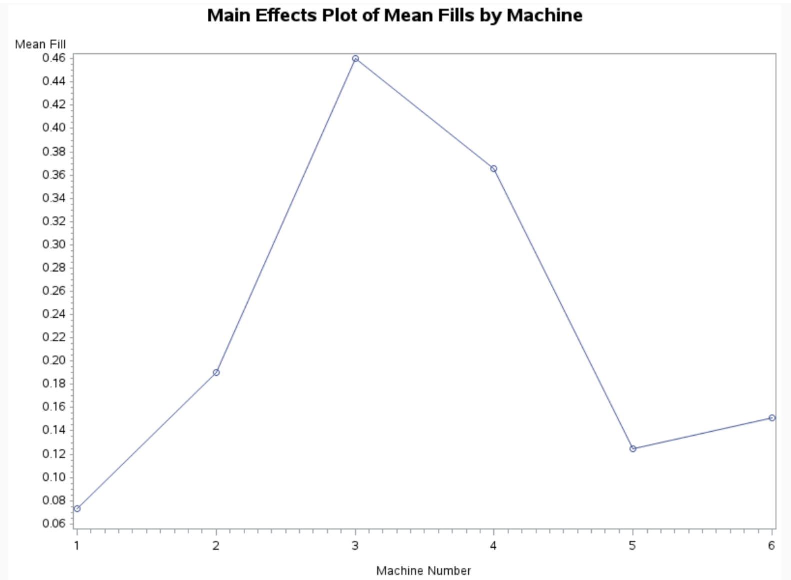
3. KNNL Problem 17.12

*17.12. Refer to **Filling machines** Problem 16.11.

a. Prepare a main effects plot of the estimated factor level means $\bar{Y}_{i\cdot}$. What does this plot suggest regarding the variation in the mean fills for the six machines?

```
proc means data=filling noprint;
    class machine;
    var fill;
    output out=meanset mean=meanfill;
run;

symbol1 v=circle i=join;
proc gplot data=meanset;
    plot meanfill*machine / vaxis=axis1 haxis=axis2;
    title 'Main Effects Plot of Mean Fills by Machine';
    axis1 label=('Mean Fill');
    axis2 label=('Machine Number');
run; quit;
```



The main effects plot of the mean fills for the six machines shows a clear variation among the machine means.
Machines 3 and 4 have noticeably higher average fills (around 0.46 and 0.36), while Machines 1 and 5 have the lowest means (around 0.07 and 0.12).

This pattern suggests that the mean fill differs substantially across machines, indicating that the factor "machine" likely has a significant effect on the average fill amount.
In other words, the filling machines are not consistent—some tend to overfill while others underfill.

b. Construct a 95 percent confidence interval for the mean fill for machine 1.

```
proc glm data=filling;
    class machine;
    model fill = machine;
    means machine / clm;
run;
```

Level of machine	N	fill Mean	fill Std Dev
1	20	0.07350000	0.19252546
2	20	0.19050000	0.18540070
3	20	0.46000000	0.16463676
4	20	0.36550000	0.16535369
5	20	0.12500000	0.17273344
6	20	0.15150000	0.17348669

Main Effects Plot of Mean Fills by Machine

The GLM Procedure

Dependent Variable: fill

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	2.28934667	0.45786933	14.78	<.0001
Error	114	3.53060000	0.03097018		
Corrected Total	119	5.81994667			

R-Square	Coeff Var	Root MSE	fill Mean
0.393362	77.29873	0.175983	0.227667

Source	DF	Type I SS	Mean Square	F Value	Pr > F
machine	5	2.28934667	0.45786933	14.78	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
machine	5	2.28934667	0.45786933	14.78	<.0001

- $\bar{Y}_1 = 0.0735$
- $MSE = 0.03097$
- $SE = \sqrt{0.03097/20} = 0.0394$
- $t_{1-\alpha/2, df_E} \cdot \sqrt{MSE/n_1} = 1.98 \cdot 0.0394 = 0.0780$

$\boxed{CI: \bar{Y}_1 \pm t_{1-\alpha/2, df_E} \cdot \sqrt{MSE/n_1}}$

$CI: 0.0735 \pm 0.0780 : (-0.0045, 0.1515)$

We are 95% confident that the true mean fill for Machine 1 lies between −0.00 and 0.15.
This again supports that Machine 1 has a lower mean fill compared with most of the other machines.

c. Obtain a 95 percent confidence interval for $D = \mu_2 - \mu_1$. Interpret your interval estimate.

```
proc glm data=filling;
    class machine;
    model fill = machine;

    /* Estimate the mean difference between Machine 2 and Machine 1 */
    estimate 'Machine2 - Machine1' machine -1 1 0 0 0 0;

run;
```

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Machine2 - Machine1	0.11700000	0.05565085	2.10	0.0377

- $t_{0.975, 114}(0.0557) = 0.110$

$CI: 0.1170 \pm \underbrace{t_{0.975,114}}_{1.98}(0.0557) \rightarrow 0.1170 \pm 0.110 : (0.01, 0.23)$

Machine 2's mean fill is significantly higher than Machine 1's by about 0.11 units on average, with the difference likely between 0.01 and 0.23 units.
Because the entire interval lies above zero, we conclude that Machine 2 fills more than Machine 1 at the 5% significance level.

KNNL Problem 17.12 For d, please calculate the confidence intervals used for the plot but ignore the plot itself.

```
proc means data=filling mean std stderr clm alpha=0.05;
    class machine;
    var fill;
run;
```

Analysis Variable : fill						
machine	N Obs	Mean	Std Dev	Std Error	Lower 95% CL for Mean	Upper 95% CL for Mean
1	20	0.0735000	0.1925255	0.0430500	-0.0166047	0.1636047
2	20	0.1905000	0.1854007	0.0414569	0.1037298	0.2772702
3	20	0.4600000	0.1646368	0.0368139	0.3829476	0.5370524
4	20	0.3655000	0.1653537	0.0369742	0.2881121	0.4428879
5	20	0.1250000	0.1727334	0.0386244	0.0441583	0.2058417
6	20	0.1515000	0.1734867	0.0387928	0.0703057	0.2326943

The intervals show that Machines 3 and 4 have substantially higher mean fills compared to Machines 1 and 5, whose intervals lie much lower.
Since many of these confidence intervals do not overlap, this suggests significant differences among the machine means — confirming that the factor "machine" affects the average fill amount.

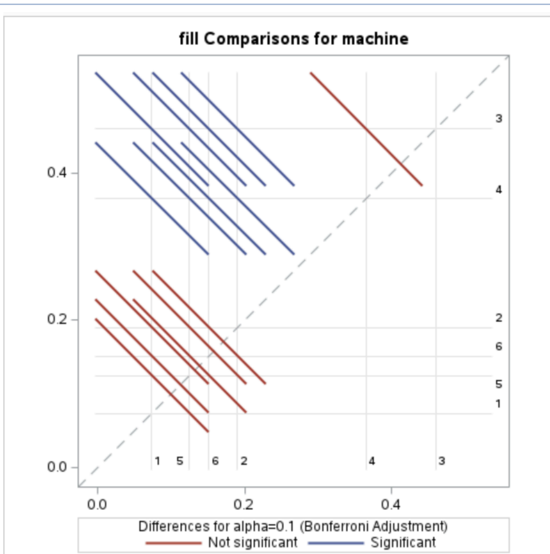
e. The consultant is particularly interested in comparing the mean fills for machines 1, 4, and 5. Use the Bonferroni testing procedure for all pairwise comparisons among these three treatment means with family level of significance $\alpha = .10$. Interpret your results and provide a graphic summary by preparing a line plot of the estimated factor level means with nonsignificant differences underlined. Do your conclusions agree with those in part (a)?

```
proc glm data=filling;
    class machine;
    model fill = machine;

    /* Perform Bonferroni multiple comparisons */
    means machine / bon alpha=0.10 cldiff;

    /* Optional: request only Machines 1, 4, and 5 for display */
    lsmeans machine / adjust=bon alpha=0.10 pdiff=all cl;
run;
quit;
```

Least Squares Means for Effect machine				
i	j	Difference Between Means	Simultaneous 90% Confidence Limits for LSMean(i)-LSMean(j)	
1	2	-0.117000	-0.270800	0.036800
1	3	-0.386500	-0.540300	-0.232700
1	4	-0.292000	-0.445800	-0.138200
1	5	-0.051500	-0.205300	0.102300
1	6	-0.078000	-0.231800	0.075800
2	3	-0.269500	-0.423300	-0.115700
2	4	-0.175000	-0.328800	-0.021200
2	5	0.065500	-0.088300	0.219300
2	6	0.039000	-0.114800	0.192800
3	4	0.094500	-0.059300	0.248300
3	5	0.335000	0.181200	0.488800
3	6	0.308500	0.154700	0.462300
4	5	0.240500	0.086700	0.394300
4	6	0.214000	0.060200	0.367800
5	6	-0.026500	-0.180300	0.127300



The Bonferroni test confirms that:
Machine 4 produces significantly higher mean fills than both Machines 1 and 5.
Machines 1 and 5 have similar mean fills.

These findings agree with the main-effects plot in part (a), where Machine 4 showed clearly higher mean fill than Machines 1 and 5.

f. Would the Tukey testing procedure have been more efficient to use in part (e) than the Bonferroni testing procedure? Explain.

No, the Tukey procedure would not have been more efficient than the Bonferroni procedure in this case.

The Tukey procedure is most efficient when all pairwise comparisons among all treatment means are of interest (here, 6 machines → 15 total comparisons).

In part (e), the consultant only wanted 3 specific pairwise comparisons (Machines 1, 4, and 5).

When only a small subset of pairwise comparisons is considered, the Bonferroni procedure is more appropriate and powerful, since it adjusts only for the few comparisons being made.

Using Tukey's method here would be too conservative, giving wider confidence intervals and making it harder to detect real differences.

4. KNNL Problem 17.17 (Bonferroni adjustment is required for part b)

Hint: you can use the following statement for $(1-\alpha)$ confidence interval of linear combination of $\sum c_i \mu_i$'s

```
proc glm data=***;
class machine;
model y=machine / clparm alpha;
estimate '**' machine c1 c2 c3 c4;
```

*17.17. Refer to **Filling machines** Problem 16.11. Machines 1 and 2 were purchased new five years ago, machines 3 and 4 were purchased in a reconditioned state five years ago, and machines 5 and 6 were purchased new last year.

a. Estimate the contrast:

$$L = \frac{\mu_1 + \mu_2}{2} - \frac{\mu_3 + \mu_4}{2}$$

```
proc mixed data=filling;
    class machine;
    model fill = machine;
    /* L = (μ1+μ2)/2 - (μ3+μ4)/2 */
    estimate 'Avg(1,2) - Avg(3,4)' machine 0.5 0.5 -0.5 -0.5 0 0 / cl alpha=0.05;
run;
quit;
```

Estimates

Label	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha	Lower	Upper
Avg(1,2) - Avg(3,4)	-0.2808	0.03935	114	-7.13	<.0001	0.05	-0.3587	-0.2028

Because the entire interval is below 0, the mean fill for Machines 1 and 2 (new five years ago) is significantly lower than the mean fill for Machines 3 and 4 (reconditioned five years ago).

In other words, Machines 3 and 4 tend to overfill more than Machines 1 and 2.

b. Estimate the following comparisons with a 90 percent family confidence coefficient; use the most efficient multiple comparison procedure:

$$D_1 = \mu_1 - \mu_2 \qquad L_1 = \frac{\mu_1 + \mu_2}{2} - \frac{\mu_3 + \mu_4}{2}$$

$$D_2 = \mu_3 - \mu_4 \qquad L_2 = \frac{\mu_1 + \mu_2}{2} - \frac{\mu_5 + \mu_6}{2}$$

$$D_3 = \mu_5 - \mu_6 \qquad L_3 = \frac{\mu_1 + \mu_2 + \mu_5 + \mu_6}{4} - \frac{\mu_3 + \mu_4}{2}$$

$$L_4 = \frac{\mu_1 + \mu_2 + \mu_3 + \mu_4}{4} - \frac{\mu_5 + \mu_6}{2}$$

Interpret your results. What can the consultant learn from these results about the differences between the six filling machines?

```
%let alphaF = 0.10;         /* family error level */
%let g      = 7;            /* number of planned comparisons */
%let aprime = %sysevalf(&alphaF/&g);  /* Bonferroni per-comparison alpha */

/* Use PROC MIXED so ESTIMATE supports /CL and alpha= */
ods output Estimates=bon_ci;    /* capture table of estimates + CIs */
```

```
proc mixed data=filling noclprint;
    class machine;
    model fill = machine;

    /* D1 = μ1 − μ2 */
    estimate 'D1: mu1 − mu2'
            machine  1 -1  0  0  0  0 / cl alpha=&aprime;

    /* D2 = μ3 − μ4 */
    estimate 'D2: mu3 − mu4'
            machine  0  0  1 -1  0  0 / cl alpha=&aprime;

    /* D3 = μ5 − μ6 */
    estimate 'D3: mu5 − mu6'
            machine  0  0  0  0  1 -1 / cl alpha=&aprime;

    /* L1 = (μ1+μ2)/2 − (μ3+μ4)/2 */
    estimate 'L1: avg(1,2) − avg(3,4)'
            machine  0.5 0.5 -0.5 -0.5 0 0 / cl alpha=&aprime;

    /* L2 = (μ1+μ2)/2 − (μ5+μ6)/2 */
    estimate 'L2: avg(1,2) − avg(5,6)'
            machine  0.5 0.5  0    0    -0.5 -0.5 / cl alpha=&aprime;

    /* L3 = (μ1+μ2+μ5+μ6)/4 − (μ3+μ4)/2 */
    estimate 'L3: avg(1,2,5,6) − avg(3,4)'
            machine  0.25 0.25 -0.5 -0.5 0.25 0.25 / cl alpha=&aprime;

    /* L4 = (μ1+μ2+μ3+μ4)/4 − (μ5+μ6)/2 */
    estimate 'L4: avg(1,2,3,4) − avg(5,6)'
            machine  0.25 0.25 0.25 0.25 -0.5 -0.5 / cl alpha=&aprime;
run;
quit;
```

Interpretation:

- The simple within-pair differences D 1 ,D 2 ,D 3 are all not significant, indicating no meaningful difference between machines purchased or reconditioned at the same time.
- The group contrasts L 1, L 3, and L 4 are significant, showing that: Machines 3 & 4 (reconditioned) have higher mean fills than Machines 1 & 2 or (1, 2, 5, 6). Machines 5 & 6 (newest) have lower mean fills than the older/ reconditioned group (1–4).

Estimates

Label	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha	Lower	Upper
D1: mu1 - mu2	-0.1170	0.05565	114	-2.10	0.0377	0.0143	-0.2555	0.02147
D2: mu3 - mu4	0.09450	0.05565	114	1.70	0.0922	0.0143	-0.04397	0.2330
D3: mu5 - mu6	-0.02650	0.05565	114	-0.48	0.6349	0.0143	-0.1650	0.1120
L1: avg(1,2) - avg(3,4)	-0.2808	0.03935	114	-7.13	<.0001	0.0143	-0.3787	-0.1828
L2: avg(1,2) - avg(5,6)	-0.00625	0.03935	114	-0.16	0.8741	0.0143	-0.1042	0.09166
L3: avg(1,2,5,6) - avg(3,4)	-0.2776	0.03408	114	-8.15	<.0001	0.0143	-0.3624	-0.1928
L4: avg(1,2,3,4) - avg(5,6)	0.1341	0.03408	114	3.94	0.0001	0.0143	0.04933	0.2189

Conclusion:

- Reconditioned Machines 3 and 4 produce significantly greater fills than all other groups.
- There are no meaningful differences between machines purchased in the same category (1 vs 2, 3 vs 4, 5 vs 6).
- Over time, the reconditioned units outperform both the oldest and the newest machines in fill volume.

5. KNNL Problem 17.25 (Bonferroni adjustment is required)

*17.25. Refer to **Filling machines** Problem 16.11. Suppose primary interest is in estimating the following comparisons:

$$L_1 = \mu_1 - \mu_2 \qquad L_3 = \frac{\mu_1 + \mu_2}{2} - \frac{\mu_3 + \mu_4}{2}$$

$$L_2 = \mu_3 - \mu_4 \qquad L_4 = \frac{\mu_1 + \mu_2 + \mu_3 + \mu_4}{4} - \frac{\mu_5 + \mu_6}{2}$$

What would be the required sample sizes if the precision of each of these comparisons is not to exceed ±.08 ounce, using the best multiple comparison procedure with a 95 percent family confidence coefficient?

```
%let mse = 0.03097;
%let alphaF = 0.05;
%let g = 4;
%let half = 0.08;

data plan;
  alpha_prime = &alphaF/&g;          /* Bonferroni per-contrast alpha */
  n = 60;                            /* candidate per-machine sample size */
  df = 6*(n-1);
  tcrit = tinv(1 - alpha_prime/2, df);

  /* Sum of coefficients squared for each contrast */
  K_L1 = 2; K_L2 = 2; K_L3 = 1; K_L4 = 0.75;

  /* Compute half-width (margin of error) for each contrast */
  me_L1 = tcrit*sqrt(&mse*K_L1/n);
  me_L2 = tcrit*sqrt(&mse*K_L2/n);
  me_L3 = tcrit*sqrt(&mse*K_L3/n);
  me_L4 = tcrit*sqrt(&mse*K_L4/n);

  put 'df=' df 5. ' tcrit=' tcrit 6.3;
  put 'Half-widths at n=60:';
  put '  L1=' me_L1 6.3 '  L2=' me_L2 6.3
      '  L3=' me_L3 6.3 '  L4=' me_L4 6.3;
run;
```

alpha_prime	n	df	tcrit
0.0125	60	354	2.5105356173

K_L1	K_L2	K_L3	K_L4
2	2	1	0.75

me_L1	me_L2	me_L3	me_L4
0.08066337	0.08066337	0.0570376159	0.0493960244

- A sample size of 60 per machine is sufficient to ensure that all four contrasts (L1–L4) have 95% family confidence intervals with precision not exceeding ± 0.08 oz.

6. KNNL Problem 18.2

18.2. A student proposed in class that deviations of the observations Y_{ij} around the estimated overall mean $\bar{Y}_{..}$ be plotted to assist in evaluating the appropriateness of ANOVA model (16.2). Would these deviations be helpful in studying the independence of the error terms? The constancy of the variance of the error terms? The normality of the error terms? Discuss.

A student suggested plotting the deviations of observations Yij around the overall mean Y.. to evaluate the appropriateness of an ANOVA model. However, this approach would not be very helpful for assessing the model assumptions because the deviations from the overall mean still contain treatment effects, not just random errors.

Independence of error terms:
Deviations from the overall mean (Yij – Y..) do not isolate the residuals. Any pattern in such a plot could be caused by treatment differences rather than dependence among errors.
Therefore, these deviations are not useful for checking independence.

Constancy of variance:
The variability of deviations around the overall mean may differ across treatments due to treatment effects, even if the true error variances are constant.
To assess homogeneity of variance, the correct approach is to plot residuals from the fitted model (eij = Yij – Yi.) against fitted values or treatment levels.
These deviations do not help in checking variance constancy.

Normality of error terms:
Because deviations from the overall mean mix group effects and random error, their distribution will not represent the actual distribution of residuals.
A normal probability plot of residuals should be used to evaluate normality.
The deviations are not appropriate for assessing normality.

Conclusion:
Plots of deviations from the overall mean (Yij – Y..) are not helpful in evaluating the independence, constancy of variance, or normality of error terms. Only residuals from the fitted ANOVA model should be used for these diagnostic checks.

7. KNNL Problem 18.17 part b to part d.

*18.17. **Winding speeds.** In a completely randomized design to study the effect of the speed of winding thread (1: slow; 2: normal; 3: fast; 4: maximum) onto 75-yard spools, 16 runs of 10,000 spools each were made at each of the four winding speeds. The response variable is the number of thread breaks during the production run. The results (in time order) are as follows:

	j						
i	1	2	3	...	14	15	16
1	4	3	2	...	2	3	4
2	7	6	4	...	4	7	6
3	12	6	14	...	13	10	14
4	17	15	7	...	19	9	23

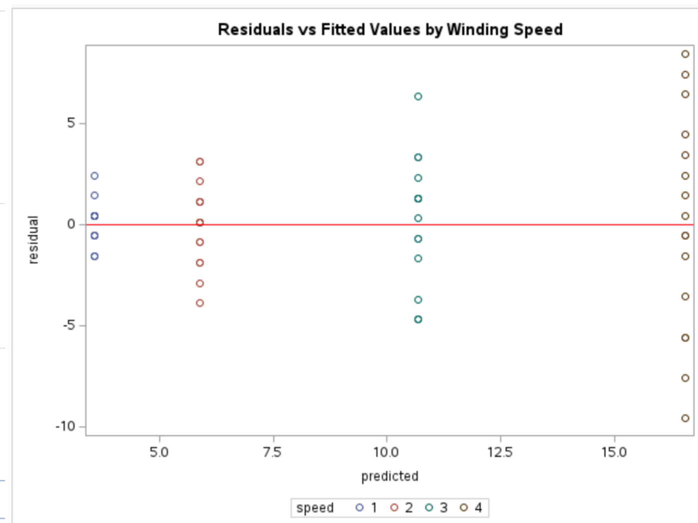
Since the responses are counts, the researcher was concerned about the normality and equal variances assumptions of ANOVA model (16.2).

b. Prepare suitable residual plots to study whether or not the error variances are equal for the four winding speeds. What are your findings?

```
proc glm data=winding plots=diagnostics;
    class speed;
    model breaks = speed;
    output out=resid p=predicted r=residual;
run;
quit;

/* Plot residuals vs fitted values and residuals vs speed */
proc sgplot data=resid;
    scatter x=predicted y=residual / group=speed;
    refline 0 / axis=y lineattrs=(color=red);
    title "Residuals vs Fitted Values by Winding Speed";
run;

proc sgplot data=resid;
    vbox residual / category=speed;
    refline 0 / axis=y lineattrs=(color=red);
    title "Boxplot of Residuals by Winding Speed";
run;
```



Findings:
The spread of residuals increases as the fitted values (and thus winding speed) increase.
Speeds 1 and 2 (slower settings) have relatively small and consistent residuals, while speeds 3 and 4 (faster settings) show much larger variation and more extreme positive and negative residuals.
This indicates nonconstant variance as the error variance grows with the winding speed.
Because the response variable (number of thread breaks) is a count, this type of variance pattern is expected.

Conclusion:
The residual plot suggests that the equal variance assumption of the ANOVA model is violated.
Variability increases with higher winding speeds. A variance-stabilizing transformation should be applied before re-analyzing the data.

c. Test by means of the Brown-Forsythe test whether or not the treatment error variances are equal; use $\alpha = .05$. What is the P-value of the test? Are your results consistent with the diagnosis in part (b)?

```
/* Brown–Forsythe HOV test */
ods output HOVFTest=bf;    /* capture the test table */
proc glm data=winding;
    class speed;
    model breaks = speed;
    means speed / hovtest=bf;
run;
quit;

/* Display p-value */
proc print data=bf noobs;
    var Source DF FValue ProbF;
    title "Brown-Forsythe Test for Equal Variances";
run;
```

The GLM Procedure

Brown and Forsythe's Test for Homogeneity of breaks Variance
ANOVA of Absolute Deviations from Group Medians

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
speed	3	111.5	37.1823	9.54	<.0001
Error	60	233.8	3.8969		

Level of speed	N	breaks Mean	breaks Std Dev
1	16	3.5625000	1.09354165
2	16	5.8750000	1.99582898
3	16	10.6875000	3.23972735
4	16	16.5625000	5.37858408

Brown-Forsythe Test for Equal Variances

Source	DF	FValue	ProbF
speed	3	9.54	<.0001
Error	60	_	_

F Value: 9.54, Pr > F: < 0.0001

At α = 0.05, we reject the null hypothesis that the error variances are equal across the four winding speeds.

Interpretation:
The extremely small P-value indicates significant differences in variability among the groups. The assumption of equal variances in the ANOVA model is violated.

Conclusion:
These results are consistent with part (b): the residual plot already suggested that variance increased with winding speed, and the Brown–Forsythe test statistically confirms this heterogeneity of variances.

d. For each winding speed, calculate $\bar{Y}_{i\cdot}$ and s_i. Examine the three relations found in the table on page 791 and determine the transformation that is most appropriate here. What do you conclude?

Means, SDs, and Transformation Guides

speed	Ybar_i	s_i	s2_over_Ybar	s_over_Ybar	s_over_Ybar2
1	3.5625	1.0935	0.3357	0.3070	0.0862
2	5.8750	1.9958	0.6780	0.3397	0.0578
3	10.6875	3.2397	0.9821	0.3031	0.0284
4	16.5625	5.3786	1.7467	0.3247	0.0196

```
/* Means and SD by speed */
proc means data=winding nway mean std;
    class speed;
    var breaks;
    output out=msd (drop=_type_ _freq_) mean=Ybar_i std=s_i;
run;

/* Three relations: s_i^2 / Ybar_i, s_i / Ybar_i, s_i / Ybar_i^2 */
data guides;
    set msd;
    s2_over_Ybar   = (s_i*s_i) / Ybar_i;
    s_over_Ybar    = s_i / Ybar_i;
    s_over_Ybar2   = s_i / (Ybar_i*Ybar_i);
run;

proc print data=guides noobs;
    var speed Ybar_i s_i s2_over_Ybar s_over_Ybar s_over_Ybar2;
    format Ybar_i s_i s2_over_Ybar s_over_Ybar s_over_Ybar2 8.4;
    title "Means, SDs, and Transformation Guides";
run;
```

Interpretation:

- $s_i^2/\bar{Y}_i$ increases with the mean, showing that the variance grows faster than the mean.
- $s_i/\bar{Y}_i$ is relatively stable (0.30–0.34).
- $s_i/\bar{Y}_i^2$ decreases as the mean increases.

The MEANS Procedure

Analysis Variable : breaks

speed	N Obs	Mean	Std Dev
1	16	3.5625000	1.0935416
2	16	5.8750000	1.9958290
3	16	10.6875000	3.2397274
4	16	16.5625000	5.3785841

Conclusion:
The most stable relation is $s_i/\bar{Y}_i$, suggesting that the logarithmic transformation (Y′ = log(Y)) is most appropriate. This transformation should stabilize the variances, making the ANOVA assumptions of equal variance and normality more reasonable.